QuickPhone — Physical ERD (22 tables)

Crow's foot notation; ANSI SQL types; one shared database for both stores. Header colour and tag = the DFD data store that holds the table.

role			D10 Staff & Stores
PK	**role_id**	SMALLINT	NN
UK	role_name	VARCHAR(20)	NN
	description	VARCHAR(200)	

store			D10 Staff & Stores
PK	**store_id**	INTEGER	NN
UK	store_name	VARCHAR(60)	NN
	address	VARCHAR(200)	NN
	phone	VARCHAR(20)	NN
	email_address	VARCHAR(120)	NN
	business_hours	VARCHAR(120)	

employee			D10 Staff & Stores
PK	**employee_id**	INTEGER	NN
FK *store*	store_id	INTEGER	NN
FK *role*	role_id	SMALLINT	NN
	first_name	VARCHAR(40)	NN
	last_name	VARCHAR(40)	NN
UK	email	VARCHAR(120)	NN
	password_hash	VARCHAR(255)	NN
	active_flag	BOOLEAN	NN
	last_login_at	TIMESTAMP	

backup_log			D11 Backup Copies
PK	**backup_id**	INTEGER	NN
	started_at	TIMESTAMP	NN
	finished_at	TIMESTAMP	
	backup_file	VARCHAR(255)	NN
	backup_status	VARCHAR(10)	NN
	restore_tested_at	TIMESTAMP	

customer			D1 Customers & Devices
PK	**customer_id**	INTEGER	NN
	first_name	VARCHAR(40)	NN
	last_name	VARCHAR(40)	NN
IX	phone	VARCHAR(20)	NN
	email	VARCHAR(120)	
	created_at	TIMESTAMP	NN

device			D1 Customers & Devices
PK	**device_id**	INTEGER	NN
FK *customer*	customer_id	INTEGER	NN
	device_type	VARCHAR(20)	NN
	brand	VARCHAR(40)	NN
	model	VARCHAR(60)	NN
IX	serial_or_imei	VARCHAR(40)	

appointment			D2 Appointments
PK	**appointment_id**	INTEGER	NN
FK *customer*	customer_id	INTEGER	NN
FK *store*	store_id	INTEGER	NN
FK *employee*	technician_id	INTEGER	NN
	scheduled_at	TIMESTAMP	NN
	appt_status	VARCHAR(12)	NN

job_status			D3 Repair Jobs
PK	**status_id**	SMALLINT	NN
UK	status_name	VARCHAR(25)	NN
	display_order	SMALLINT	NN

repair_job			D3 Repair Jobs
PK	**repair_job_id**	INTEGER	NN
UK	ticket_number	VARCHAR(12)	NN
FK *device*	device_id	INTEGER	NN
FK *store*	store_id	INTEGER	NN
FK,UK *appt.*	appointment_id	INTEGER	
FK *employee*	intake_employee_id	INTEGER	NN
FK *employee*	assigned_tech_id	INTEGER	NN
FK *job_status*	current_status_id	SMALLINT	NN
	reported_issue	VARCHAR(500)	NN
	intake_condition	VARCHAR(500)	
	check_in_at	TIMESTAMP	NN
	final_cost	DECIMAL(10,2)	
	completed_at	TIMESTAMP	
	picked_up_at	TIMESTAMP	
FK *employee*	released_by_id	INTEGER	
	id_verified_flag	BOOLEAN	NN
	signature_file	VARCHAR(255)	

job_status_history			D3 Repair Jobs
PK	**status_history_id**	BIGINT	NN
FK *repair_job*	repair_job_id	INTEGER	NN
FK *job_status*	status_id	SMALLINT	NN
FK *employee*	changed_by_id	INTEGER	NN
	changed_at	TIMESTAMP	NN
	change_note	VARCHAR(300)	

notification			D6 Notifications
PK	**notification_id**	INTEGER	NN
FK *repair_job*	repair_job_id	INTEGER	NN
FK *employee*	sent_by_id	INTEGER	NN
	recipient_email	VARCHAR(120)	NN
	delivery_status	VARCHAR(10)	NN
	sent_at	TIMESTAMP	NN
	failure_reason	VARCHAR(300)	

quote			D4 Quotes
PK	**quote_id**	INTEGER	NN
FK *repair_job*	repair_job_id	INTEGER	NN
FK *employee*	prepared_by_id	INTEGER	NN
	labor_amount	DECIMAL(10,2)	NN
	parts_amount	DECIMAL(10,2)	NN
	total_amount	DECIMAL(10,2)	NN
	customer_decision	VARCHAR(10)	NN
	decision_at	TIMESTAMP	
FK *employee*	decision_by_id	INTEGER	
	created_at	TIMESTAMP	NN

payment			D5 Payments
PK	**payment_id**	INTEGER	NN
FK *repair_job*	repair_job_id	INTEGER	NN
FK *employee*	processed_by_id	INTEGER	NN
	amount	DECIMAL(10,2)	NN
	payment_method	VARCHAR(12)	NN
	payment_status	VARCHAR(10)	NN
	paid_at	TIMESTAMP	NN

job_note			D3 Repair Jobs
PK	**job_note_id**	INTEGER	NN
FK *repair_job*	repair_job_id	INTEGER	NN
FK *employee*	employee_id	INTEGER	NN
	note_type	VARCHAR(20)	NN
	note_text	VARCHAR(2000)	NN
	created_at	TIMESTAMP	NN

job_part			D3 Repair Jobs
PK	**job_part_id**	INTEGER	NN
FK *repair_job*	repair_job_id	INTEGER	NN
FK *part*	part_id	INTEGER	NN
FK *store*	store_id	INTEGER	NN
	quantity	INTEGER	NN
	part_status	VARCHAR(8)	NN
FK *employee*	recorded_by_id	INTEGER	NN
	recorded_at	TIMESTAMP	NN

part			D7 Parts Inventory
PK	**part_id**	INTEGER	NN
FK *supplier*	default_supplier_id	INTEGER	
	part_name	VARCHAR(100)	NN
	device_model	VARCHAR(60)	
	grade	VARCHAR(20)	
	standard_cost	DECIMAL(10,2)	NN
	active_flag	BOOLEAN	NN

store_part_stock			D7 Parts Inventory
PK,FK *store*	**store_id**	INTEGER	NN
PK,FK *part*	**part_id**	INTEGER	NN
	quantity_on_hand	INTEGER	NN
	reorder_point	INTEGER	NN

inventory_txn			D7 Parts Inventory
PK	**inventory_txn_id**	BIGINT	NN
FK *stock*	store_id	INTEGER	NN
FK *stock*	part_id	INTEGER	NN
FK *repair_job*	repair_job_id	INTEGER	
FK *purch._order*	purchase_order_id	INTEGER	
FK *employee*	recorded_by_id	INTEGER	NN
	txn_type	VARCHAR(12)	NN
	quantity_change	INTEGER	NN
	reason	VARCHAR(200)	
	txn_at	TIMESTAMP	NN

supplier			D9 Suppliers
PK	**supplier_id**	INTEGER	NN
UK	supplier_name	VARCHAR(100)	NN
	contact_name	VARCHAR(80)	
	phone	VARCHAR(20)	
	email	VARCHAR(120)	
	active_flag	BOOLEAN	NN

purchase_order			D8 Purchase Orders
PK	**purchase_order_id**	INTEGER	NN
FK *store*	store_id	INTEGER	NN
FK *supplier*	supplier_id	INTEGER	NN
FK *employee*	created_by_id	INTEGER	NN
FK *employee*	approved_by_id	INTEGER	
IX	po_status	VARCHAR(20)	NN
	created_at	TIMESTAMP	NN
	submitted_at	TIMESTAMP	
	approved_at	TIMESTAMP	
	placed_at	TIMESTAMP	
	rejection_reason	VARCHAR(300)	
	order_total	DECIMAL(10,2)	NN

po_line			D8 Purchase Orders
PK	**po_line_id**	INTEGER	NN
FK *purch._order*	purchase_order_id	INTEGER	NN
FK *part*	part_id	INTEGER	NN
	quantity_ordered	INTEGER	NN
	quantity_received	INTEGER	NN
	unit_cost	DECIMAL(10,2)	NN

po_job_link			D8 Purchase Orders
PK,FK *purch._order*	**purchase_order_id**	INTEGER	NN
PK,FK *repair_job*	**repair_job_id**	INTEGER	NN

Notation

exactly one → zero or many

exactly one → one or many

zero or one → zero or one

PK	primary key (shaded row)
FK table	foreign key and the table it references
UK / IX	unique constraint / non-unique index
NN	NOT NULL; blank = nullable

FKs to store and employee are listed in each key column; their lines are left off so the diagram stays readable (see the relationship list in the data dictionary).

DFD data store holding each entity

D1 **Customers & Devices:** customer, device
D2 **Appointments:** appointment
D3 **Repair Jobs:** job_status, repair_job, job_status_history, job_note, job_part
D4 **Quotes:** quote
D5 **Payments:** payment
D6 **Notifications:** notification
D7 **Parts Inventory:** part, store_part_stock, inventory_txn
D8 **Purchase Orders:** purchase_order, po_line, po_job_link
D9 **Suppliers:** supplier
D10 **Staff & Stores:** role, store, employee
D11 **Backup Copies:** backup_log

Physical design decisions

1. One relational database shared by both stores, behind the web app; ANSI SQL types (DBMS not yet chosen).
2. backup_log supports the nightly backup and restore test (NFR06). employee.password_hash is a salted hash; sign-in sessions are held by the web framework, not in the data model.
3. Stored derived values kept on purpose: quote.total_amount, purchase_order.order_total, repair_job.current_status_id.
4. CHECKs: quantity_on_hand >= 0 (FR07); job_part.quantity > 0; 0 <= quantity_received <= quantity_ordered; inventory_txn.reason required for Adjustment; status/type columns limited to the Sprint 2 value lists.
5. Indexes beyond keys: customer.phone, device.serial_or_imei, repair_job (assigned_tech_id, current_status_id), repair_job (store_id, current_status_id), purchase_order.po_status, appointment (store_id, scheduled_at).
6. Unique keys: repair_job.ticket_number, repair_job.appointment_id, employee.email; po_line (purchase_order_id, part_id).
7. Composite keys: store_part_stock (store_id, part_id), po_job_link (purchase_order_id, repair_job_id).